

Homework #2

Math 104: Intro to Analysis

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1 Problem 1

Let (X, d) be a metric space. A subset $E \subset X$ is called *dense* in E if, for any open $O \subset X$, we have $O \cap E \neq \emptyset$. A metric space (X, d) is called *separable* if it has a countable dense subset.

- (a) Show that $\mathbb{Q}$ is dense in $\mathbb{R}$, and conclude that $\mathbb{R}$ is separable.
- (b) Show that a subset E is dense in X if and only if $\overline{E} = X$.
- (c) By following these steps, prove that any compact metric space (X, d) is separable:
 - (i) Using problem 5(c) on homework 2, show that for each $n \in \mathbb{N}$, the following is true: there is a finite set $S_n \subset X$ so that every point of X is within distance $\frac{1}{n}$ of some point of S_n .
 - (ii) Let $S = \bigcup_{n=1}^{\infty} S_n$. Show that S is countable.
 - (iii) Show that every point of X is either in S or a limit point of S . Conclude that S is dense in X , and hence that X is separable.

1.1 Solution (a)

Alexis's Proposition: For any $a, b \in \mathbb{R}$ where $a < b$, there exists some $\frac{m}{n} \in \mathbb{Q}$ such that $a < \frac{m}{n} < b$.

Proof. Let $a, b \in \mathbb{R}$ where $a < b$. Since $a < b$, $b - a > 0$. By the Archimedean property, there exists some $n \in \mathbb{N}$ such that $n(b - a) > 1$. Distribute and add na to both sides to find $nb > na + 1$. Also by the Archimedean property, there exists some natural numbers m_1 and m_2 such that $m_1 > na$ and $m_2 > -na$. The latter means that $na > -m_2$. Therefore, $m_1 > na > -m_2$. Since m_1 and $-m_2$ are both integers, there exists some integer m such that $-m_2 < m < m_1$ and $m - 1 \leq na < m$. This is equivalent to $m \leq na + 1 < m + 1$. Combine all of these and we find that $na < m \leq na + 1 < nb$. Shortening this, $na < m < nb$. Divide this all by n to find $a < \frac{m}{n} < b$. Since n is a natural number and m is an integer, $\frac{m}{n}$ is a rational number. Ergo, if $a, b \in \mathbb{R}$ and $a < b$, there exists some $\frac{m}{n} \in \mathbb{Q}$ such that $a < \frac{m}{n} < b$. OEΔ

Now we prove $\mathbb{Q}$ is dense in $\mathbb{R}$ with the usual metric.

Proof. Take any set O open in $\mathbb{R}$. This means that for any $o \in O$, there exists some $r > 0$ such that there exists an open ball $B_r(o)$ that is a subset of O . We define this open ball as $B_r(o) = \left\{ x \in \mathbb{R} \mid |x - o| < r \right\}$.

Since $r > 0$, this means $r > -r$ and therefore $o + r > o - r$. By Alexis's Proposition, there exists some rational number $\frac{m}{n}$ such that $o - r < \frac{m}{n} < o + r$. Subtract o from all sides to find $-r < \frac{m}{n} - o < r$. By the definition of the absolute value, $\left| \frac{m}{n} - o \right| < r$. Therefore, $\frac{m}{n} \in B_r(o)$ and by extension $\frac{m}{n} \in O$. Therefore, $\frac{m}{n} \in \mathbb{Q} \cap O$. Therefore, $\mathbb{Q} \cap O \neq \emptyset$. This means that for any open $O \subset \mathbb{R}$, $\mathbb{Q} \cap O \neq \emptyset$. Ergo, $\mathbb{Q}$ is dense in $\mathbb{R}$. OEΔ

Last we prove that $\mathbb{R}$ is separable.

Proof. We know that $\mathbb{Q}$ is dense in $\mathbb{R}$. We know that $\mathbb{Q}$ is a subset of $\mathbb{R}$. We know that $\mathbb{Q}$ is countable. Therefore, $\mathbb{Q}$ is a countable, dense subset of $\mathbb{R}$. Ergo, $\mathbb{R}$ is separable. OEΔ

1.2 Solution (b)

First, we prove that if E is dense in X , then $\overline{E} = X$.

Proof. Suppose that E is dense in X . Take any point $x \in X$. Create an open ball of radius r for any $r > 0$. We can define the open ball as $B_r(x) = \{y \in X \mid |y - x| < r\}$. Since open balls are open and E is dense in X , there is a point $p \in E \cap B_r(x)$. Therefore, there is a point p in E in every open ball centered at any point in X . Therefore, every point in X is a limit point of E . Therefore, X is the set of limit points of E . Since $\overline{E}$ is the union of E and the set of its limit points and every point in E is in X (since E is a subset of X), $X = \overline{E}$. OEΔ

Next we prove that E is dense in X if $\overline{E} = X$.

Proof. Let $\overline{E} = X$. Take any open set $O \subset X$. Since $\overline{E} = E \cup E'$, for any point $p \in X$, either $p \in E$ or $p \in E'$.

If $p \in E$, then $p \in E \cap O$, so $E \cap O \neq \emptyset$.

If $p \in E'$, then p is a limit point of E . Since O is open, there is some open ball of radius r centered around p such that $B_r(p) \subset O$. Since p is a limit point of E , there exists some $q \in E$ such that $p \neq q$ and $q \in B_r(p) \subset O$. Therefore, since $q \in E$ and $q \in O$, $q \in O \cap E$. Therefore, $E \cap O \neq \emptyset$.

Ergo, if $\overline{E} = X$, E is dense in X . OEΔ

1.3 Solution (c)

1.3.1 Part (i)

Proof. We know that X is compact. From Problem 5c on Homework 2, we know that for a compact set has the following property: for all $\varepsilon > 0$, there is a finite set $\{x_1, \dots, x_k\} \subset X$ such that each $y \in X$ is within distance ε of some x_j . Suppose that $\varepsilon = \frac{1}{n}$ for some $n \in \mathbb{N}$. Label the finite set for this value of n $\{x_1, \dots, x_k\}$ as S_n . Ergo, for any $n \in \mathbb{N}$, there is a finite set $S_n \subset X$ so that every point of X is within distance $\frac{1}{n}$ of some point in S_n . OEΔ

1.3.2 Part (ii)

Proof. We know that for any $n \in \mathbb{N}$, S_n is finite. We also know that the union of at most countable sets of at most countable cardinality is at most countable. Therefore, since we are working with countable sets that are finite, the union of S_n for all $n \in \mathbb{N}$ is countable. Ergo, $S = \bigcup_{n=1}^{\infty} S_n$ is countable. OEΔ

1.3.3 Part (iii)

Proof. We have two cases here: for any $x \in X$, either $x \in S$ or $x \notin S$. If $x \in S$, then we don't need anything further.

Suppose instead that $x \notin S$. We proved earlier that every point $x \in X$ is within distance $\frac{1}{n}$ of some point $s \in S$ for any $n \in \mathbb{N}$, or in other words $d(x, s) < \frac{1}{n}$. For x to be a limit point of S , it must be that for any $r > 0$, there is a point $y \in S$ such that $y \neq x$ and $y \in B_r(x)$ for the open ball of radius r centered at x . $B_r(x)$ is defined as $B_r(x) = \{z \in X \mid d(x, z) < r\}$. The Archimedean property proved that for any $a > 0$, there exists some natural number n such that $a \geq \frac{1}{n} > 0$. Therefore, for any $r > 0$, there exists some natural number n such that $r \geq \frac{1}{n} > 0$. Furthermore, we know from earlier that for every point $x \in X$, there is some point $s \in S$ such that $d(x, s) < \frac{1}{n}$ for any $n \in \mathbb{N}$. Transistively, this means that for any $r > 0$ and for any $x \in X$, there is some $s \in S$ such that $d(x, s) < r$. Since $x \notin S$, $x \neq s$, so $d(x, s) > 0$. Therefore, $s \in B_r(x)$ and $s \in S$ for any $r > 0$ and for any $x \in X \setminus S$. Therefore, x is a limit point of S .

Ergo, every point of X is either in S or is a limit point of S . This means that $X = \overline{S}$. This means that S is dense in X by part (ii). Since S is countable and a metric space is countable if it has a dense subset, (X, d) is separable. OEΔ

2 Problem 2

Using **only** the definitions of $\lim$, $\limsup$, and $\liminf$, prove the following statements.

- (a) $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = 0$. *Hint: Use difference of squares.*
- (b) $\limsup_{n \rightarrow \infty} (-2)^n = \infty$.
- (c) $\limsup_{n \rightarrow \infty} [-\sqrt{n}] = -\infty$.
- (d) $\liminf_{n \rightarrow \infty} \left[\frac{1}{n} + (1 + (-1)^n)n \right] = 0$.

2.1 Solution (a)

First we prove that for all $m, n \in \mathbb{N}$, if $m \geq n$, then $\frac{1}{2\sqrt{m}} < \frac{1}{2\sqrt{n}}$.

Proof. To begin, we know that $n \leq m$ for $m, n \in \mathbb{N}$. Take the square root of both sides to find that $\sqrt{n} \leq \sqrt{m}$ ¹. Double both sides to find $2\sqrt{n} \leq 2\sqrt{m}$. Divide each side by what is on the other side to find $\frac{1}{2\sqrt{m}} < \frac{1}{2\sqrt{n}}$. OEΔ

By a similar logic, we prove that for all $n \in \mathbb{N}$, $\frac{1}{\sqrt{n+1}+\sqrt{n}} < \frac{1}{2\sqrt{n}}$.

Proof. To begin, we know that $n < n+1$ for $n \in \mathbb{N}$. Take the square root of both sides to find that $\sqrt{n} < \sqrt{n+1}$ ². Add $\sqrt{n}$ to both sides to find $2\sqrt{n} < \sqrt{n+1} + \sqrt{n}$. Divide each side by what is on the other side to find $\frac{1}{\sqrt{n+1}+\sqrt{n}} < \frac{1}{2\sqrt{n}}$. OEΔ

Proof. We can start by rewriting the term within the limit.

$$\sqrt{n+1} - \sqrt{n} = (\sqrt{n+1} - \sqrt{n}) \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{n+1-n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \quad (1)$$

Let $\varepsilon > 0$. Choose a natural number N such that $N > \frac{1}{4\varepsilon^2}$ and let $n \in \mathbb{N}$ such that $n \geq N$. Take our inequality and multiply by ε^2 while dividing by N to find that $\varepsilon^2 > \frac{1}{4N}$. Take the square root of both sides to find $\varepsilon > \frac{1}{2\sqrt{N}}$. Based on earlier statements and recalling $n \geq N$, we find $\frac{1}{2\sqrt{N}} \geq \frac{1}{2\sqrt{n}}$. Also based on earlier, $\frac{1}{2\sqrt{n}} > \frac{1}{\sqrt{n+1}+\sqrt{n}}$. Transistively, this means $\varepsilon > \frac{1}{\sqrt{n+1}+\sqrt{n}}$.

Since $n \in \mathbb{N}$, $n > 0$, and the same can be said about $n+1$. Therefore, $\sqrt{n} > 0$ and $\sqrt{n+1} > 0$. Collectively adding, that means $\sqrt{n+1} + \sqrt{n} > 0$. Multiplying everything by $(\sqrt{n+1} + \sqrt{n})^{-1}$ twice, we find that $\frac{1}{\sqrt{n+1}+\sqrt{n}} > 0$. Therefore, $\left| \frac{1}{\sqrt{n+1}+\sqrt{n}} \right| = \frac{1}{\sqrt{n+1}+\sqrt{n}}$.

We can therefore say that $\varepsilon > \left| \frac{1}{\sqrt{n+1}+\sqrt{n}} \right|$ for any $\varepsilon > 0$ and for any $n \geq N$ for some $N \in \mathbb{N}$. Therefore, $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}+\sqrt{n}} = 0$. Since $\frac{1}{\sqrt{n+1}+\sqrt{n}} = \sqrt{n+1} - \sqrt{n}$, $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}+\sqrt{n}}$. Ergo, $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = 0$. OEΔ

¹Cite a source.

²Cite a source.

2.2 Solution (b)

2.3 Solution (c)

2.4 Solution (d)

3 Problem 3

Let $\{s_n\}$ be an arbitrary sequence in $\mathbb{R}$.

(a) Let $s \in \mathbb{R}$. Show that $\limsup_{n \rightarrow \infty} s_n = s$ if and only if the following two conditions hold:

- For all $\epsilon > 0$, there is $N \in \mathbb{N}$ so that $s_n < s + \epsilon$ for all $n \geq N$.
- For all $\epsilon > 0$ and for all $N \in \mathbb{N}$, there exists $n \geq N$ so that $s_n > s - \epsilon$.

(b) Show that some subsequence of $\{s_n\}$ converges (in the sense of extended real numbers, if necessary) to $\limsup_{n \rightarrow \infty} s_n$ and some subsequence of $\{s_n\}$ converges to $\liminf_{n \rightarrow \infty} s_n$.

(c) Define

$$E = \{x \in [-\infty, \infty] : \text{some subsequence } \{s_{n_k}\} \text{ of } \{s_n\} \text{ converges to } x\}.$$

Show that $\limsup_{n \rightarrow \infty} s_n = \sup E$ and $\liminf_{n \rightarrow \infty} s_n = \inf E$.

Hint: what happens to $\limsup$ and $\liminf$ on subsequences?

3.1 Solution (a)

First we prove the forward direction.

Proof. Suppose that $\limsup_{n \rightarrow \infty} s_n = s$.

OEΔ

3.2 Solution (b)

3.3 Solution (c)

4 Problem 4

Let $a > 1$. In this problem, we will define what it means to write a^r for a positive *irrational* number r . We first define exponentiation for positive *integer* exponents in the usual way, via repeated multiplication. We know this satisfies some properties, for example

$$a^{r+s} = a^r a^s \quad (*)$$

$$(ab)^r = a^r b^r \quad (\Delta)$$

for $a, b > 0$ and $r, s > 0$ integers. Now, we proved in class that for any positive integer q , there is a *unique* positive number whose q th power is a , which we denote $\sqrt[q]{a}$. For a positive rational number $r = \frac{p}{q}$, we define $a^r = \sqrt[q]{a^p}$.

- Show that this definition makes sense, i.e. is independent of the representation of r as a fraction. That is, show that if rational numbers $\frac{p}{q} = \frac{r}{s}$ then $\sqrt[q]{a^p} = \sqrt[s]{a^r}$. *Hint:* use lowest terms and the uniqueness of q th roots.
- Show that properties $(*)$ and (Δ) above still hold for positive rational numbers. *Hint:* find a common denominator and use uniqueness again.
- For positive rational $r < s < t$, show that $a^r < a^s$ and that

$$0 < a^s - a^r = a^r (a^{s-r} - 1).$$

Conclude that for any positive rational r, s and any rational t larger than both r and s , we have

$$|a^r - a^s| < a^t (a^{|r-s|} - 1).$$

- Now let r be a positive *irrational* number. We know there is a sequence of rational numbers q_k that converge to r (for example, the decimal expansion of r). We define

$$a^r = \lim_{k \rightarrow \infty} a^{q_k}.$$

Show that the sequence $\{a^{q_k}\}$ is Cauchy, and conclude that this limit exists.

Hint: Use part (c) and Rudin Theorem 3.20(d).

- Show that the choice of sequence $\{q_k\}$ is irrelevant; that is, show that if $\{p_k\}$ and $\{q_k\}$ are sequences of rational numbers converging to r , then

$$\lim_{k \rightarrow \infty} a^{p_k} = \lim_{k \rightarrow \infty} a^{q_k}.$$

- Using limit laws, prove that properties $(*)$ and (Δ) continue to hold for positive irrational exponents.

Remark: For $r < 0$ we then define $a^r = \frac{1}{a^{-r}}$, and for $0 < a < 1$ we define $a^r = \left(\frac{1}{a}\right)^{-r}$. It is then a matter of checking cases to verify that all the properties of exponentiation we want are indeed true.

4.1 Solution (a)

Billie's Proposition: If $\frac{p}{q} = \frac{r}{s}$ and $\frac{p}{q}$ is in lowest terms but $\frac{r}{s}$ is not, there exists some nonzero integer t such that $r = pt$ and $s = qt$.

Proof. Let $\frac{p}{q} = \frac{r}{s}$ and $\frac{p}{q}$ is in lowest terms but $\frac{r}{s}$ is not. Multiply by to find that $ps = qr$.

Divide by p to find that $s = \frac{qr}{p}$. Since s is a nonzero integer, p divides qr . Since $\frac{p}{q}$ is in lowest terms, p does not divide q , so p must divide r . Therefore, there is some nonzero integer t such that $p = rt$.

Substitute in our value of p into the result of the cross multiplication to find $rst = qr$. Divide both sides by r to find that $q = st$. OEΔ

Proof. Let $x = \sqrt[q]{a^p}$ and $y = \sqrt[s]{a^r}$, and suppose that $\frac{p}{q} = \frac{r}{s}$ and that $\frac{p}{q}$ is in lowest terms while $\frac{r}{s}$ is not. This would mean that $x = a^{\frac{p}{q}}$ and $y = a^{\frac{r}{s}}$. Take x to the power of q and y to the power of s to find $x^q = a^p$ and $y^s = a^r$. Take the former to the power of r and the latter to the power of p to find $x^{qr} = a^{pr}$ and $y^{ps} = a^{pr}$. Transistively, this means $x^{qr} = y^{sp}$.

By Billie's Proposition, since $\frac{p}{q}$ is in lowest terms while $\frac{r}{s}$ is not, there exists some nonzero integer t such that $r = pt$ and $s = qt$. Substitute these into our equation to find that $x^{qpt} = y^{qpt}$.

$$x^{qr} = y^{sp} \tag{2}$$

$$x = y \tag{3}$$

$$(x^{qr})^{sp} = (y^{sp})^{qr} \tag{4}$$

$$x^{qr} = (y^{sp})^{\frac{q}{p} \cdot \frac{r}{s}} \tag{5}$$

OEΔ

4.2 Solution (b)

4.3 Solution (c)

4.4 Solution (d)

4.5 Solution (e)

4.6 Solution (f)